Contact, Power, and Balls in Play: A Three-Rate Decomposition of Batting Average and the Disappearance of the .400 Hitter

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Abstract

Batting average factors exactly as BA $= c\{h + (1-h)\,b\}$, where c is the rate of avoiding a strikeout, h is the home-run rate among non-strikeout at-bats, and b is batting average on balls in play (BABIP). The bracketed term is the rate f at which non-strikeout at-bats become hits, so the decomposition nests the two-factor identity BA $= cf$. Using the 256 major-league hitters with at least 300 at-bats in 2025, every season from 2004 to 2025, and 20,357 qualifying seasons from 1901 to 2019, we ask which rates behave like skills and how the corresponding abilities are related. Over 19 consecutive-season pairs the median year-to-year correlation is 0.86 for c, 0.74 for h and 0.41 for b; batting average (0.44) is no more repeatable than BABIP. A trivariate logistic-normal random-effects model gives single-season reliabilities of 0.91, 0.76 and 0.30, and a talent correlation between contact and power of -0.67 (95% profile interval -0.75 to -0.57), while BABIP talent is only weakly related to either. Historically, the negative covariance between c and f, which was near zero in the dead-ball era, explains why batting averages have become less dispersed, and so why no one has hit .400 since 1941. That covariance runs almost entirely through home runs: removing the contact–power channel from the 2010s covariance recovers 72% of the gap between modern and dead-ball dispersion in batting-average talent.

Keywords: batting average; BABIP; empirical Bayes; multivariate random effects; reliability; shrinkage; .400 hitter

1 Introduction

Batting average is the oldest summary of hitting and one of the most studied quantities in sports statistics. It served as the running example in the empirical Bayes work of Efron and Morris (1975, 1977), and Brown (2008) used it to compare empirical Bayes and Bayes predictors within a season. These analyses treat a hitter's batting average as a single binomial proportion whose true value is estimated by shrinking the observed average toward a common mean (Morris, 1983). Batting average is notoriously unstable from season to season, and a large literature, popular and academic, attributes much of that instability to chance (Albert and Bennett, 2001).

Batting average is, however, a composite. An at-bat ends in a strikeout or it does not; a non-strikeout at-bat is a home run or a ball in play; and a ball in play is a hit or an out. These three events depend on the hitter in different ways. Avoiding strikeouts reflects pitch recognition and bat-to-ball skill. Home runs need no help from the defense and depend mostly on how hard and at what angle the ball is struck. Whether a ball in play falls for a hit depends also on fielders, parks and chance. McCracken (2001) argued that pitchers have little control over hits on balls in play, and the same reasoning underlies the use of batting average on balls in play (BABIP) as an indicator of hitter luck (Tango et al., 2007).

Several authors have used a partial version of this structure. Bickel (2004) wrote batting average as the product of one minus the strikeout rate and the in-play average to explain why .400 has become so hard to reach. Albert (2005) fitted beta-binomial random-effects models to several hitting rates and found that strikeout and home-run rates are far more ability-driven than batting average, and Albert (2016) showed that predicting component rates separately and recombining them improves forecasts of aggregate measures. What has not been examined is the joint structure of the components: how the underlying abilities are correlated once sampling noise is removed, and how that correlation has changed over the history of the game.

This paper writes batting average exactly as the product of three rates and asks three questions:

1. Which rates behave like stable skills, and how much of each rate's cross-sectional variation reflects ability rather than chance (Sections 4 and 5)?

2. How are the three abilities related once chance is removed, and what does the dependence imply for estimating an individual hitter's ability (Section 5)?

3. Gould (1986, 1996) attributed the disappearance of the .400 hitter to a narrowing of the distribution of batting averages. Which components drove that narrowing (Section 6)?

The answers are that strikeout avoidance and home-run hitting are skills while BABIP is mostly noise; that contact and power abilities are strongly negatively correlated; and that the emergence of this contact–power trade-off, rather than a convergence of hitting talent, accounts for most of the reduced dispersion of modern batting averages.

2 Data and the decomposition

2.1 The identity

For a hitter with AB at-bats, SO strikeouts, HR home runs and H hits, define

$$c = \frac{AB - SO}{AB}, \qquad h = \frac{HR}{AB - SO}, \qquad b = \frac{H - HR}{AB - SO - HR}. \tag{1}$$

We call c the contact rate, h the home-run rate and b BABIP. Our b differs from the conventional BABIP only in omitting sacrifice flies from the denominator. Because $ch = HR/AB$ and $c(1-h)b =$

Table 1: Four 2025 hitters with similar batting averages and different combinations of contact rate c, home-run rate h and BABIP b.

Player	AB	H	SO	HR	BA	c	h	b
Aaron Judge	541	179	160	53	0.331	0.704	0.139	0.384
Miguel Andujar	321	102	49	10	0.318	0.847	0.037	0.351
Jonathan Aranda	370	117	107	14	0.316	0.711	0.053	0.414
Jacob Wilson	486	151	39	13	0.311	0.920	0.029	0.318

$(H - HR)/AB$,

$$\text{BA} = \frac{H}{AB} = c\,\{h + (1-h)\,b\} = c\,f, \qquad f = \frac{H}{AB - SO} = h + (1-h)\,b, \tag{2}$$

exactly, for every hitter and season. The finishing rate f is the proportion of non-strikeout at-bats that become hits. Writing $\text{BA} = cf$ gives the coarser two-factor view used by Bickel (2004); splitting f into h and b separates the part of finishing that the defense cannot affect from the part it can. Among the 2025 hitters the identity holds to within 1e-16; 3 of them hit no home runs.

As an illustration, Aaron Judge had 541 at-bats in 2025 and struck out 160 times, so $c = 381/541 = 0.704$. Of the 381 non-strikeout at-bats, 53 were home runs ($h = 0.139$), and 126 of the remaining 328 balls in play fell for hits ($b = 0.384$). His 179 hits in 541 at-bats give BA = 0.331.

2.2 Samples

All data come from the Lahman database (Friendly et al., 2026), analyzed in R (R Core Team, 2024). Stints with different teams in a season are combined. The main sample consists of the 256 hitters with at least 300 at-bats in 2025. For repeatability we use every season from 2004 to 2025 with the same threshold. For the historical analysis we use the 20,357 American, National and Federal League player-seasons from 1901 to 2019 with at least 300 at-bats and a recorded strikeout total; the Negro Leagues, recently added to the database, are excluded because their strikeout records are incomplete. Seasons are grouped by decade.

3 The cross-section in 2025

3.1 Equal averages, different compositions

Table 1 shows four 2025 hitters whose averages fall between .311 and .331. Judge and Jacob Wilson reached similar averages by opposite routes: Judge with a low contact rate, a home-run rate nearly five times Wilson's and a high BABIP; Wilson with the second-highest contact rate among qualifiers and little power. Jonathan Aranda had the highest home-run rate, and Miguel Andujar the highest BABIP, among the other hitters within twenty points of Judge's average.

Figure 1 shows the pairwise relationships among the three rates for all 256 hitters. Contact and home-run rates are strongly negatively correlated ($r = -0.53$, 95% interval -0.61 to -0.43). BABIP

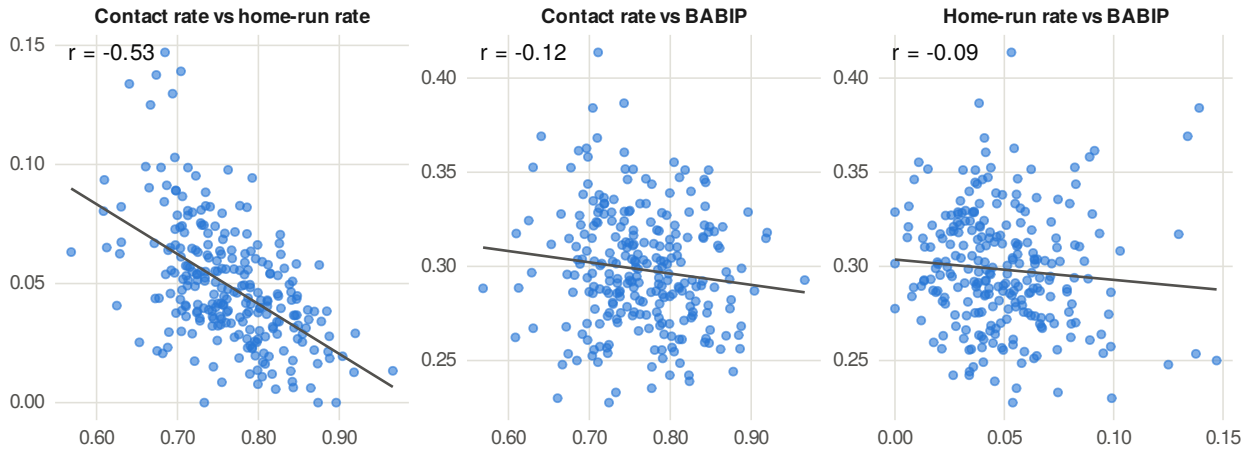


Figure 1: Pairwise relationships among contact rate c, home-run rate h and BABIP b for the 256 hitters with at least 300 at-bats in 2025. Lines are least-squares fits; r is the Pearson correlation.

is only weakly related to contact ($r = -0.12$) and to home-run rate ($r = -0.09$). At the two-factor level the correlation between c and f is -0.40 (-0.50 to -0.29). The familiar observation that hitters who rarely strike out do less damage when they make contact is therefore, in 2025, primarily a statement about home runs.

3.2 A variance decomposition

Taking logarithms of (2) gives $\log \mathrm{BA} = \log c + \log f$, so the cross-sectional variance decomposes exactly as

$$\mathrm{Var}(\log \mathrm{BA}) = \mathrm{Var}(\log c) + \mathrm{Var}(\log f) + 2\,\mathrm{Cov}(\log c, \log f). \tag{3}$$

In 2025 the three terms are 0.0072, 0.0095 and -0.0066, summing to 0.0102. The two factors contribute comparable variance, but because they are negatively correlated the observed dispersion of batting average is 39% smaller than it would be if c and f varied independently (0.0167). This covariance term is the quantity that drives the historical argument of Section 6.

Because f is a sum rather than a product, $\mathrm{Var}(\log f)$ cannot be split exactly. Linearizing about the means $\bar{h}$, $\bar{b}$ and $\bar{f}$,

$$\mathrm{Var}(\log f) \approx \frac{1}{\bar{f}^2}\Big\{(1-\bar{b})^2\,\mathrm{Var}(h) + (1-\bar{h})^2\,\mathrm{Var}(b) + 2(1-\bar{b})(1-\bar{h})\,\mathrm{Cov}(h,b)\Big\}. \tag{4}$$

In 2025 the three terms are 0.0029, 0.0080 and -0.0008, shares of 29%, 80% and -8%; their sum is within 5.2% of the exact $\mathrm{Var}(\log f)$. On observed rates, then, BABIP accounts for most of the variation in finishing. Much of that variation is sampling noise, however. Subtracting the average binomial variance $\overline{p(1-p)/n}$ from each observed variance removes 66% of the variance of b but only 21% of that of h, after which the home-run and BABIP terms account for 55% and 65% of the total. Once chance is removed, power and BABIP contribute about equally to differences in

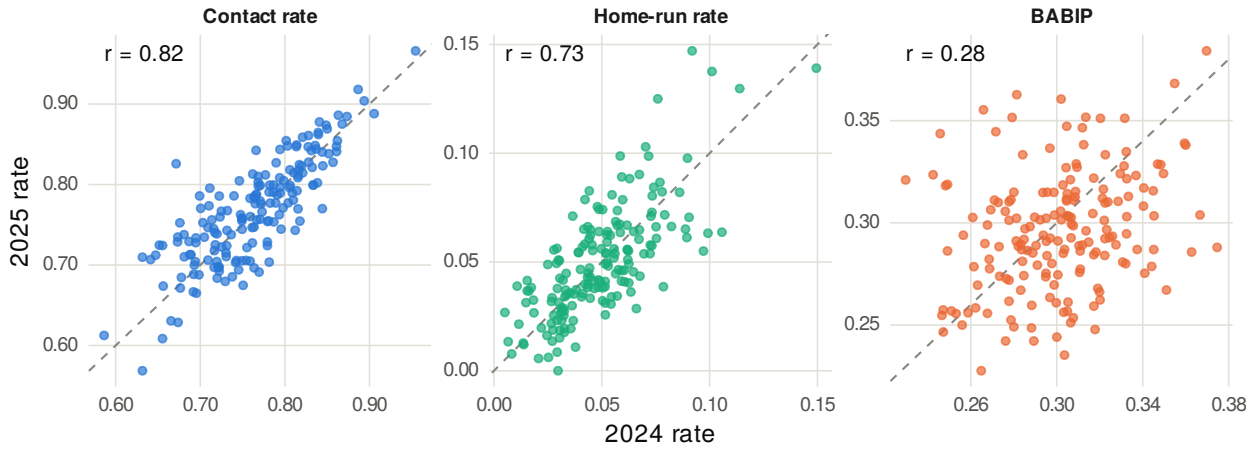


Figure 2: 2024 and 2025 rates for the 176 hitters with at least 300 at-bats in both seasons. The dashed line is $y = x$.

finishing ability.

4 Which rates are skills?

A rate that measures ability should be repeatable. For each pair of consecutive seasons from 2004 to 2025 we correlate each rate across the hitters who qualified in both seasons. The 2020 season, in which almost no one reached 300 at-bats, drops out, leaving 19 pairs with a median of 183 hitters each.

Figure 2 shows the 2024–2025 pair (176 hitters). Contact and home-run rates repeat closely ($r = 0.82$ and 0.73); BABIP barely does ($r = 0.28$). Figure 3 shows that the pattern holds in every pair. The year-to-year correlation ranges from 0.81 to 0.88 for c (median 0.86), from 0.66 to 0.80 for h (median 0.74), and from 0.24 to 0.52 for b (median 0.41). For batting average it ranges from 0.30 to 0.53 (median 0.44): the composite is no more repeatable than its noisiest component.

5 A trivariate random-effects model

5.1 Model and estimation

Repeatability uses two seasons. A random-effects model extracts the same information from a single season and also estimates how the three abilities are related. Because the three outcomes are nested, each can be modeled as binomial conditional on the one before it. For hitter i,

$$\begin{aligned} AB_i - SO_i &\sim \mathrm{Bin}(AB_i,\, c_i), & \mathrm{logit}(c_i) &= \mu_c + u_i, \\ HR_i &\sim \mathrm{Bin}(AB_i - SO_i,\, h_i), & \mathrm{logit}(h_i) &= \mu_h + v_i, \\ H_i - HR_i &\sim \mathrm{Bin}(AB_i - SO_i - HR_i,\, b_i), & \mathrm{logit}(b_i) &= \mu_b + w_i, \end{aligned} \tag{5}$$

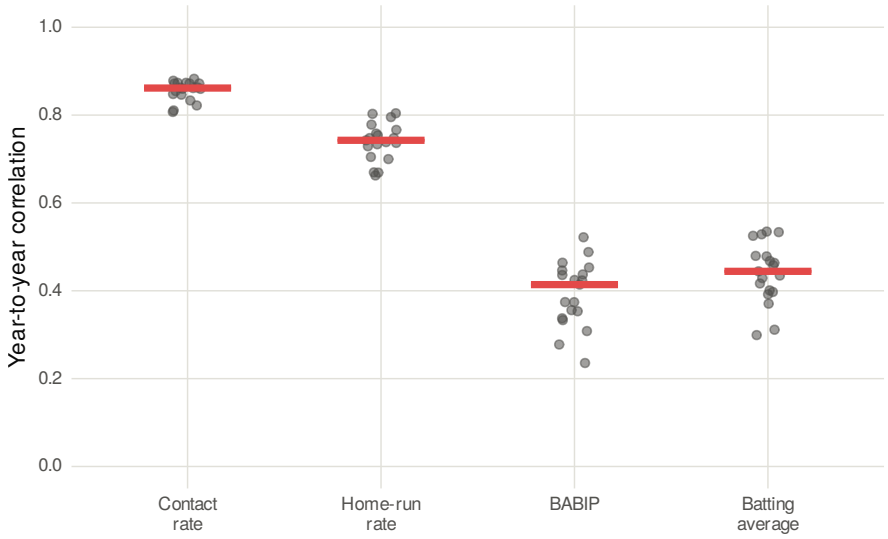


Figure 3: Year-to-year correlations for each of the 19 consecutive-season pairs from 2004 to 2025. Horizontal bars mark the medians.

Table 2: Maximum likelihood estimates for model (5), 2025 season (logit scale). Wald intervals for the means; profile-likelihood intervals for the standard deviations and correlations.

Parameter	Interpretation	Estimate	95% interval
μ_c	league contact rate (0.771)	1.211	(1.164, 1.258)
μ_h	league home-run rate (0.045)	-3.059	(-3.128, -2.989)
μ_b	league BABIP (0.299)	-0.853	(-0.870, -0.835)
σ_c	talent SD, contact	0.369	(0.335, 0.407)
σ_h	talent SD, home runs	0.494	(0.439, 0.556)
σ_b	talent SD, BABIP	0.078	(0.053, 0.100)
ρ_{ch}	talent correlation, contact and home runs	-0.67	(-0.75, -0.57)
ρ_{cb}	talent correlation, contact and BABIP	-0.22	(-0.45, 0.00)
ρ_{hb}	talent correlation, home runs and BABIP	-0.23	(-0.48, 0.02)

with $(u_i, v_i, w_i)^\top \sim \mathrm{N}_3(\mathbf{0}, \Sigma)$ and Σ unstructured, with standard deviations $\sigma_c, \sigma_h, \sigma_b$ and correlations $\rho_{ch}, \rho_{cb}, \rho_{hb}$. Given the abilities, the three binomial outcomes are conditionally independent because they are determined by disjoint sets of at-bats. We fit (5) by maximum likelihood with the Laplace approximation in `lme4` (Bates et al., 2015), stacking the three outcomes and specifying an unstructured covariance for the three random intercepts, and compute profile-likelihood intervals for the variance components.

Table 2 gives the estimates for 2025. The means correspond to league rates of 0.771 for contact, 0.045 for home runs and 0.299 for BABIP. On the logit scale the spread of talent is largest for home runs ($\sigma_h = 0.494$), intermediate for contact ($\sigma_c = 0.369$), and small for BABIP ($\sigma_b = 0.078$).

5.2 Talent correlations

The estimated correlation between contact and home-run abilities is $\hat{\rho}_{ch} = -0.67$, noticeably stronger than the raw correlation of -0.53. Sampling noise inflates the variance of each observed rate without changing their covariance, so observed correlations are attenuated toward zero (Spearman, 1904); the model removes that attenuation. Covariances between observed rates need no such correction, because the three rates are computed from disjoint sets of at-bats and their sampling errors are independent. BABIP ability is only weakly related to the other two abilities ($\hat{\rho}_{cb} = -0.22$, $\hat{\rho}_{hb} = -0.23$), and neither interval clearly excludes zero.

5.3 Single-season reliability

By the delta method, the sampling variance of hitter i's observed logit for a rate p based on n trials is approximately $\tau_i^2 = 1/\{n\,\hat{p}(1-\hat{p})\}$; we add $\frac{1}{2}$ to success and failure counts so that a season with no home runs remains finite. The ratio $w_i = \sigma^2/(\sigma^2 + \tau_i^2)$ is the fraction of between-hitter variation in that rate that reflects ability, the reliability of classical test theory (Lord and Novick, 1968). Averaged over the 2025 hitters the reliability is 0.91 for contact, 0.76 for home runs and 0.30 for BABIP. The mean noise standard deviations are 0.114, 0.285 and 0.121: home-run rates are noisier than BABIP in absolute terms, but their talent spread is more than six times larger. These single-season reliabilities agree closely with the multi-season correlations of Section 4, as they should, since reliability and test–retest correlation estimate the same quantity when ability is stable.

5.4 Trivariate shrinkage

Let z_i hold hitter i's observed logits minus (μ_c, μ_h, μ_b) and let $T_i = \mathrm{diag}(\tau_{i,c}^2, \tau_{i,h}^2, \tau_{i,b}^2)$. Under a normal approximation to the binomial likelihood on the logit scale, the posterior mean of his abilities is

$$\hat{\theta}_i = \Sigma\,(\Sigma + T_i)^{-1} z_i. \tag{6}$$

When Σ is diagonal, (6) reduces to shrinking each rate separately by its own reliability, the familiar James–Stein form (Efron and Morris, 1975). When it is not, the off-diagonal terms let each observed rate inform the estimates of the other abilities.

Table 3 and Figure 4 show the conditional modes from the fitted model, which use the full binomial likelihood. Contact and home-run rates barely move; the mean absolute adjustments are 0.005 and 0.005. BABIP moves much more (mean adjustment 0.018): Judge's observed .384 becomes an estimated ability of 0.308, close to the league mean of 0.299, and Aranda's .414 becomes 0.326. The correlations matter for individual hitters. Shrinking Judge's BABIP by its own reliability alone gives 0.325, whereas (6) gives 0.309, because his high home-run rate, through $\hat{\rho}_{hb} < 0$, points to a lower BABIP ability. Averaged over hitters, the correlations shift the BABIP estimate by 0.004.

Table 3: Observed 2025 rates and model estimates (conditional modes) for the four hitters of Table 1.

	Contact rate		Home-run rate		BABIP	
Player	observed	estimate	observed	estimate	observed	estimate
Aaron Judge	0.704	0.694	0.139	0.118	0.384	0.308
Miguel Andujar	0.847	0.833	0.037	0.032	0.351	0.305
Jonathan Aranda	0.711	0.712	0.053	0.049	0.414	0.326
Jacob Wilson	0.920	0.899	0.029	0.023	0.318	0.294

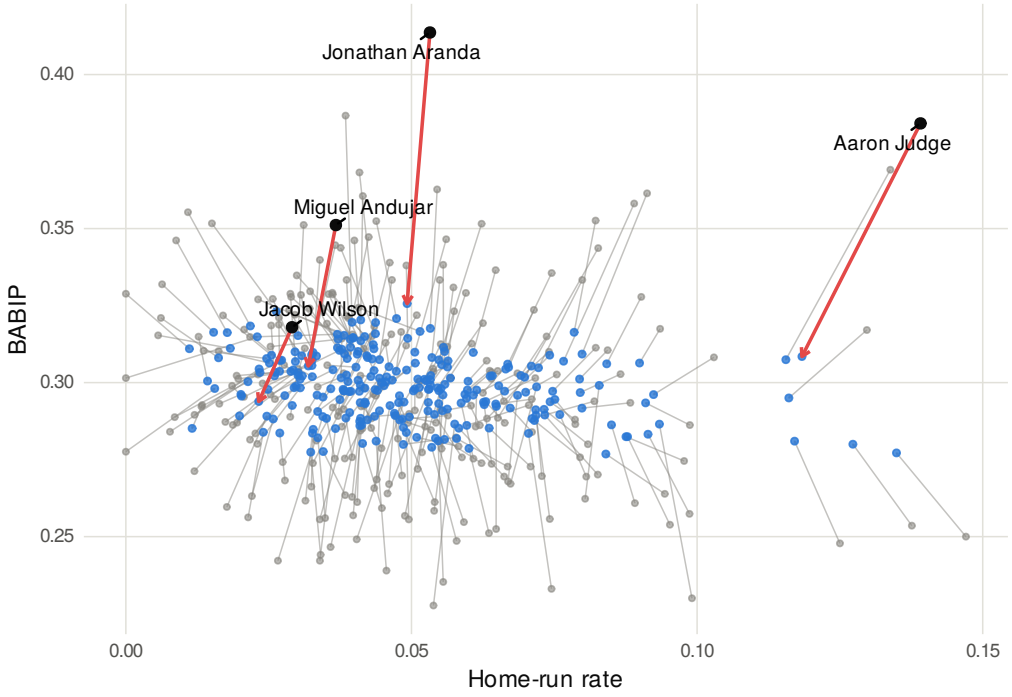


Figure 4: Observed 2025 home-run rate and BABIP (gray) and model estimates (blue) for the 256 qualifiers. Arrows run from observed to estimated; the four hitters of Table 1 are highlighted. The arrows are nearly vertical: BABIP is shrunk strongly toward the league mean, home-run rate much less.

6 A historical perspective on the .400 hitter

6.1 Gould's argument and the two-factor decomposition

Among American, National and Federal League hitters with at least 300 at-bats there have been 13 seasons of .400 or better since 1901, the last in 1941. Gould (1986, 1996) argued that .400 disappeared not because hitters became worse but because the distribution of batting averages narrowed: as play improved and the talent pool deepened, the distance between ordinary and exceptional hitters shrank, trimming both tails. The standard deviation of batting average among qualifiers did fall, from 0.037 in the 1900s to 0.030 in the 1940s (about 19%), and has been nearly constant since (0.029 in the 2010s; Figure 5a).

Decomposition (3) shows where the narrowing came from. Table 4 reports its terms by decade,

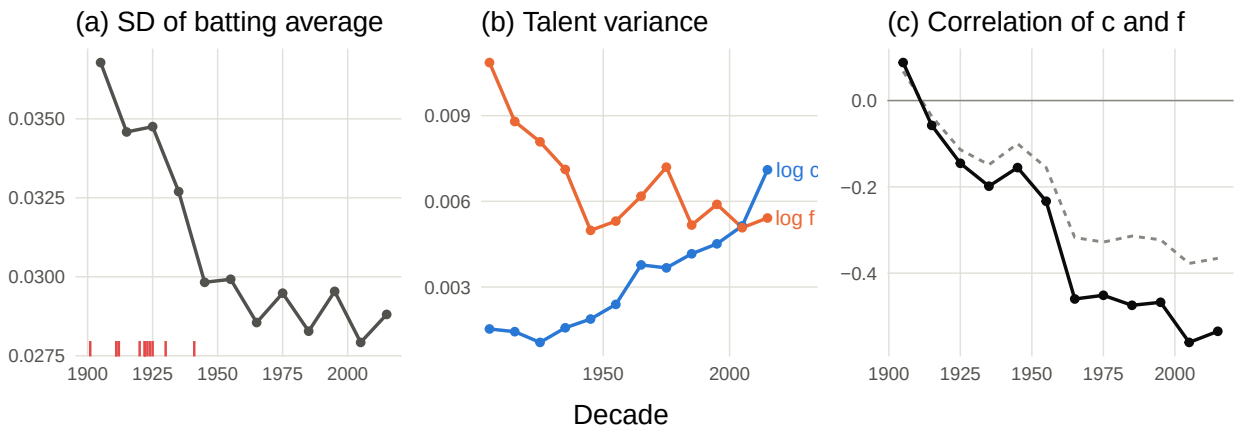


Figure 5: Qualifying AL, NL and FL hitters by decade, 1900s–2010s. (a) Standard deviation of batting average; ticks mark the 13 seasons of .400 or better. (b) Talent-scale variances of $\log c$ and $\log f$ (observed variances minus average sampling variance). (c) Correlation between c and f, observed (dashed) and with sampling noise removed (solid).

on the observed scale and, after subtracting the average delta-method sampling variance $(1-p)/(np)$ from $\mathrm{Var}(\log c)$ and $\mathrm{Var}(\log f)$, on the talent scale. Sampling noise matters most for f, accounting for 36% of the observed variance of $\log f$ in the 1900s and 51% in the 2010s. Three trends emerge (Figure 5b,c).

1. Strikeout avoidance did not homogenize. Its talent variance on the log scale grew by a factor of about 4.6 from the 1900s to the 2010s.

2. The talent variance of finishing roughly halved (ratio 0.50), consistent with Gould's picture of a converging talent pool.

3. The two factors were essentially uncorrelated in the dead-ball era (talent correlation 0.09), while since the 1960s the observed correlation has stayed between -0.38 and -0.31 (talent correlation -0.53 in the 2010s).

A counterfactual makes the role of the covariance precise. On the talent scale, the variance of $\log \mathrm{BA}$ fell from 0.0131 in the 1900s to 0.0059 in the 2010s. Combining the 2010s talent variances of $\log c$ and $\log f$ with the 1900s covariance gives 0.0132, essentially the dead-ball level. The growth in the spread of contact ability and the narrowing of finishing ability roughly offset, and the reduced spread of modern batting averages is due almost entirely to the stronger negative covariance between the two factors. Gould's account is therefore incomplete: a .400 season requires excellence in both c and f at once, and the modern game rarely produces that combination.

6.2 The trade-off runs through home runs

The three-rate decomposition identifies which part of finishing trades off against contact. Figure 6 and Table 5 show the correlations among c, h and b by decade. The correlation between contact and home runs was close to zero in the 1900s (-0.04), when home runs were rare, fell to -0.40 in

Table 4: Decomposition (3) by decade for qualifying hitters. Columns 3–8 are observed quantities; the last three remove binomial sampling noise from the variances of $\log c$ and $\log f$ and from their correlation.

								Talent scale		
Decade	n	SD(BA)	Var($\log c$)	Var($\log f$)	2Cov	Var($\log$ BA)	$r(c,f)$	Var($\log c$)	Var($\log f$)	$r(c,f)$
1900s	1105	0.037	0.0018	0.0170	0.0007	0.0195	0.07	0.0015	0.0109	0.09
1910s	1276	0.035	0.0017	0.0147	-0.0004	0.0160	-0.04	0.0014	0.0088	-0.06
1920s	1241	0.035	0.0012	0.0132	-0.0009	0.0136	-0.11	0.0011	0.0081	-0.15
1930s	1296	0.033	0.0018	0.0123	-0.0013	0.0128	-0.15	0.0016	0.0071	-0.20
1940s	1211	0.030	0.0021	0.0107	-0.0010	0.0119	-0.10	0.0019	0.0050	-0.16
1950s	1227	0.030	0.0027	0.0111	-0.0017	0.0121	-0.16	0.0024	0.0053	-0.23
1960s	1581	0.029	0.0042	0.0121	-0.0044	0.0118	-0.32	0.0038	0.0062	-0.46
1970s	2038	0.029	0.0040	0.0130	-0.0046	0.0124	-0.33	0.0037	0.0072	-0.45
1980s	2114	0.028	0.0046	0.0110	-0.0044	0.0112	-0.31	0.0042	0.0052	-0.47
1990s	2229	0.030	0.0050	0.0115	-0.0048	0.0116	-0.32	0.0045	0.0059	-0.47
2000s	2545	0.028	0.0056	0.0105	-0.0057	0.0104	-0.38	0.0051	0.0051	-0.56
2010s	2494	0.029	0.0077	0.0111	-0.0066	0.0122	-0.37	0.0071	0.0054	-0.53

Table 5: Correlations among the three rates by decade for qualifying hitters. $\bar{h}$ is the mean home-run rate. Columns headed r are observed correlations; "talent r" columns divide the observed covariance by noise-corrected standard deviations. The last column is the home-run term's share of approximation (4).

Decade	n	SD(BA)	$\bar{h}$	$r(c,f)$	$r(c,h)$	$r(c,b)$	$r(h,b)$	talent $r(c,h)$	talent $r(c,b)$	HR share
1900s	1105	0.037	0.005	0.07	-0.04	0.08	0.35	-0.06	0.10	1%
1920s	1241	0.035	0.014	-0.11	-0.40	0.01	0.30	-0.46	0.01	10%
1940s	1211	0.030	0.020	-0.10	-0.45	0.10	-0.01	-0.51	0.16	18%
1960s	1581	0.029	0.033	-0.32	-0.49	-0.09	0.05	-0.56	-0.15	24%
1980s	2114	0.028	0.031	-0.31	-0.58	-0.03	-0.09	-0.66	-0.05	24%
2000s	2545	0.028	0.042	-0.38	-0.54	-0.11	0.00	-0.62	-0.19	27%
2010s	2494	0.029	0.045	-0.37	-0.58	-0.08	-0.09	-0.67	-0.13	25%

the 1920s as the home run entered the game, and reached -0.58 by the 2010s. The correlation between contact and BABIP was small in every decade. Hitters who struck out more did not do systematically better or worse on balls in play; they hit more home runs.

Linearizing f makes the attribution explicit:

$$\mathrm{Cov}(c,f) \approx (1-\bar{b})\,\mathrm{Cov}(c,h) + (1-\bar{h})\,\mathrm{Cov}(c,b). \tag{7}$$

The home-run term accounts for 76% of $\mathrm{Cov}(c,f)$ in the 1960s and 80% in the 2010s, and for more than all of it from the 1920s through the 1950s, when the BABIP term was slightly positive.

6.3 Home runs became part of finishing

Figure 7 applies approximation (4) by decade. In the 1900s the average qualifying hitter homered in about one of every 200 non-strikeout at-bats ($\bar{h} = 0.005$), and home runs accounted for only 1% of the variation in finishing rates; finishing was essentially BABIP. By the 2010s the mean home-run rate was about nine times higher ($\bar{h} = 0.045$), and home runs accounted for 25% of that variation (40% once sampling noise is removed). Over the century the talent variance of h

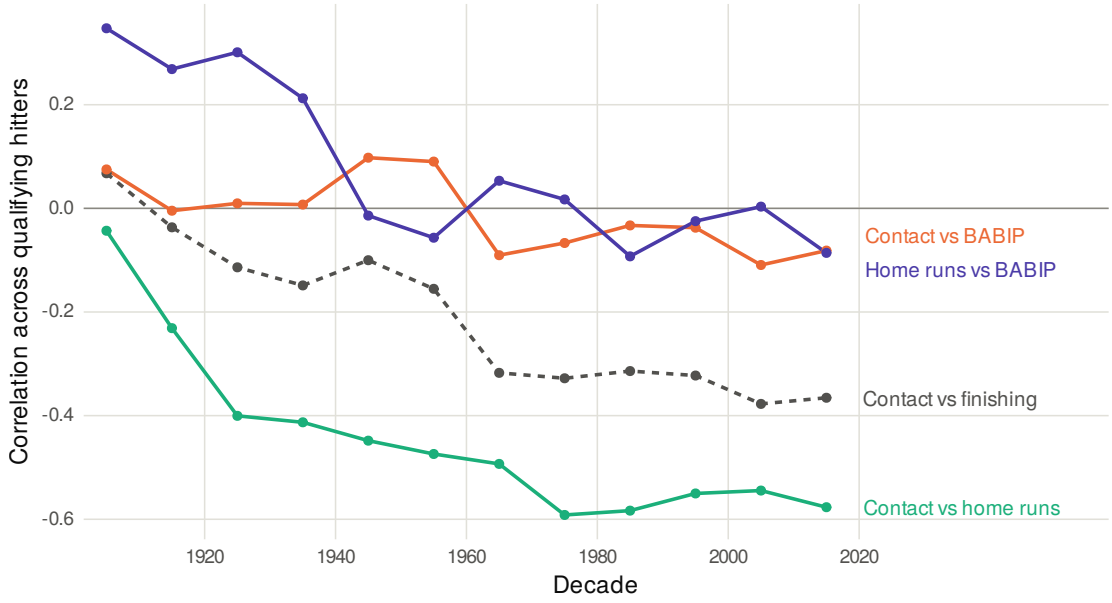


Figure 6: Observed correlations among the three rates, and between contact and finishing (dashed), for qualifying hitters in each decade, 1900s–2010s.

grew about 33-fold, while that of b fell to 0.64 of its 1900s level. Power became a second, growing dimension along which hitters differ, and it is the dimension that trades off against contact.

6.4 How much did the contact–power trade-off matter?

Equation (7) supports a narrower counterfactual than the one above: keep everything about the 2010s except the contact–home-run channel. Dividing (7) by $\bar{c}\bar{f}$ approximates $\mathrm{Cov}(\log c, \log f)$; for the 2010s it gives -0.0033, matching the exact value of -0.0033. Setting the home-run term to zero leaves -0.0007, and the talent-scale variance of $\log \mathrm{BA}$ becomes 0.0112. That is 72% of the way from the actual 2010s value (0.0059) to the value with dead-ball covariance (0.0132). Roughly three-quarters of the reduced dispersion of modern batting-average talent is attributable to the trade-off between avoiding strikeouts and hitting home runs.

7 Discussion

The identity $\mathrm{BA} = c\{h + (1-h)b\}$ separates batting average into two components that behave like skills and one that behaves largely like noise. Strikeout avoidance and home-run rate are highly repeatable and highly reliable within a single season; BABIP is neither, and it is the main reason batting average is so unstable. Because the identity is exact, the variance decompositions and counterfactuals of Sections 3 and 6 require no modeling assumptions beyond the linearization in (4) and (7), whose accuracy we check in the Appendix. The random-effects model adds estimates of the talent correlations and a trivariate shrinkage estimator that lets each rate inform the

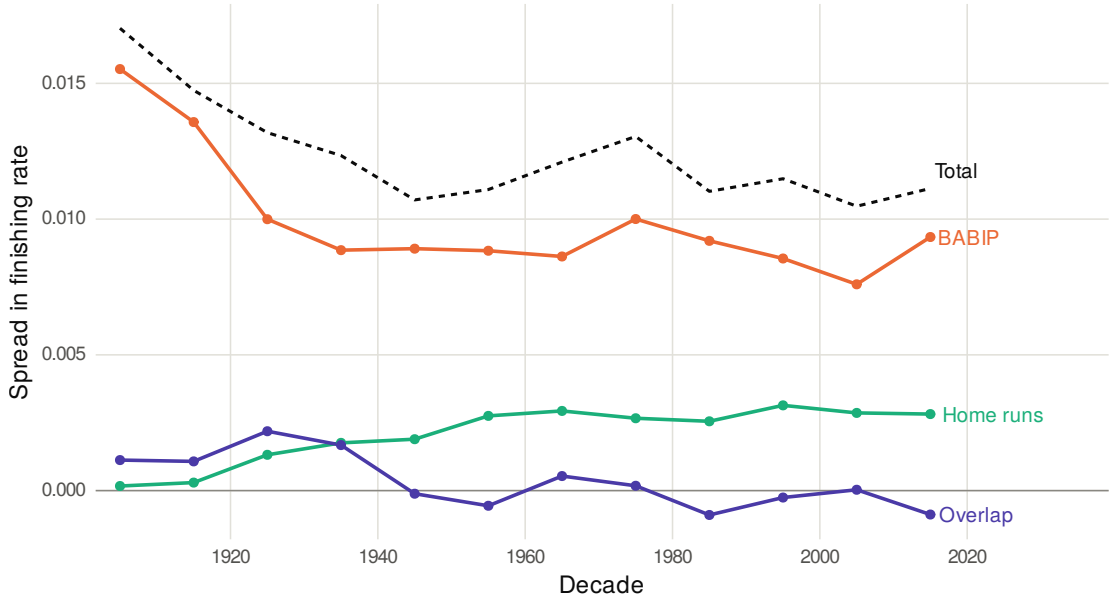


Figure 7: Approximation (4) of $\mathrm{Var}(\log f)$ by decade: the home-run term, the BABIP term and the covariance (overlap) term. The dashed line is the exact $\mathrm{Var}(\log f)$.

others, extending the component approach of Albert (2016) by modeling the dependence between components.

The historical analysis refines Gould's explanation for the disappearance of the .400 hitter. Batting averages did become less dispersed, but not because hitters became more alike: the spread of contact ability grew substantially. What changed is that, beginning in the 1920s, the hitters who make the most contact have tended to hit the fewest home runs. A modern .400 season would require a hitter who rarely strikes out, hits for power, and enjoys a favorable season on balls in play, a combination that has been rare since the home run became central to the game.

Several limitations should be noted. First, BABIP reflects both contact quality and defense; batted-ball tracking data could separate the two. Second, home-run rates depend on parks and on the baseball itself, so part of what we call power is environmental. Third, at-bats exclude walks, hit-by-pitches and sacrifices, so the rates describe the at-bat outcomes that enter batting average rather than all plate appearances. Fourth, the 300-at-bat threshold selects hitters whose teams kept playing them, which may truncate the lower tails of the rates. Fifth, the model is fit to one season at a time; a model with player effects shared across seasons, or a fully Bayesian fit with priors on Σ (Gelman and Hill, 2007), would give sharper estimates and full posterior uncertainty for the correlations. Finally, the historical analysis establishes when the contact–power trade-off emerged, not why. Changes in pitching, in hitters' approaches, and in how teams value strikeouts relative to power are all candidates.

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The analyses and this manuscript were produced in collaboration with Claude (Anthropic), a large language model. The author posed the research questions, chose which directions to pursue, and reviewed the results. Claude wrote the R code, carried out the computations, drafted the text, and assembled the references. Claude is not listed as an author because it cannot take responsibility for the work; the author takes full responsibility for the content, including any remaining errors. A single R script reproduces every number, table and figure and is available from the author.

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A Accuracy of the linear approximations

Approximation (4) replaces $\mathrm{Var}(\log f)$ by a first-order expansion of $\log\{h + (1-h)b\}$ about $(\bar{h}, \bar{b})$. In 2025 the approximation is within 5.2% of the exact value. Across the twelve decades from the 1900s to the 2010s the relative error has median 0.9% and maximum 2.4%. Approximation (7), divided by $\bar{c}\bar{f}$, reproduces the exact 2010s value of $\mathrm{Cov}(\log c, \log f)$ to four decimal places (-0.0033 versus -0.0033).

Noise corrections subtract the average binomial sampling variance from each observed variance: $\overline{p(1-p)/n}$ on the rate scale, with $n = AB$ for c, $AB - SO$ for h and $AB - SO - HR$ for b, and $\overline{(1-p)/(np)}$ on the log scale. Covariances are not corrected, because the three rates are computed from disjoint sets of at-bats and their sampling errors are independent. "Talent" correlations in Tables 4 and 5 divide the observed covariance by the noise-corrected standard deviations.